Algebraik ifodalar

Qutida nechta konfet borligini bilmasak ham, uni harf bilan yozib qo'yamiz.

1-QISM

Mavzu

ESLAB OLAYLIK

- $\square$ o'rnidagi sonni topish: $7 + \square = 25, \square = 25 - 7 = 18; \square : 6 = 7, \square = 7 \cdot 6 = 42$.
- $\square$ o'rniga harf ham yozish mumkin: $7 + x = 25$.
- To'g'ri to'rtburchakning yuzi $=$ bo'yi $\cdot$ eni, bo'yi $=$ yuzi $:$ eni.

1-QOIDA · HARF BILAN BELGILASH

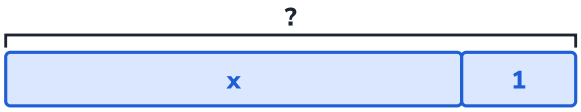
Biz algebrada noma'lum sonni ifodalash uchun harflardan foydalanamiz.

Javobni qavs ichida yozish kerak. Biz x ning qiymatini bilmaymiz, shuning uchun javob $(x + 1)$.

$x + 1, x + 2$ va $x + 3$ **algebraik ifodalarga** misol bo'ladi.

1-MISOL · LIFT

Liftda x kishi bor edi. Yana 1 kishi kirdi. Liftda necha kishi bo'ldi?



Bu rasm — chizma-model: x va 1 birga

1. Agar $x = 5$ bo'lsa, $5 + 1 = 6$ kishi bo'lardi.
2. x ni bilmaymiz, javob ifoda bo'lib qoladi: $(x + 1)$ kishi.

2-MISOL · AVTOBUS

Avtobusda 30 ta yo'lovchi bor edi. Bekatda n kishi tushdi. Avtobusda nechta yo'lovchi qoldi?



1. Tushganlar ayriladi: $30 - n$.
2. Avtobusda $(30 - n)$ ta yo'lovchi qoldi.



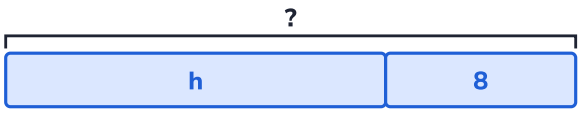
Demak, x o'rniga istalgan sonni qo'ysa bo'lar ekan-da?

USUL · JUMLADAN IFODAGA

Jumla	Algebraik ifoda
m ga 15 ni qo'shing	$m + 15$ yoki $15 + m$
h dan 8 ga ortiq (8 ta ko'p)	$h + 8$
40 dan k ni ayiring	$40 - k$
r dan 9 ga kam	$r - 9$

1-MISOL · ORTIQ

«h dan 8 ga ortiq» ni ifoda bilan yozing.



1. h ga yana 8 qo'shiladi.
2. Natija: $h + 8$.

2-MISOL · AYIRISH

«40 dan k ni ayiring» ni ifoda bilan yozing.



1. Butun tasma 40, undan k ajratiladi.
2. To'g'ri yozuv $40 - k$, $k - 40$ emas.

«4 marta ko'p» bilan «4 ta ko'p» bir xilmi?



2-QOIDA · KO'PAYTIRISH

$5 \cdot q = q \cdot 5$.

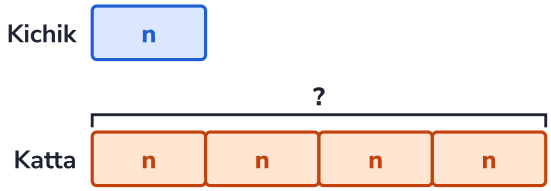
Biz $5 \cdot q$ ni $5q$ kabi yozamiz.

$1 \cdot q = q \cdot 1$.

$1 \cdot q$ ni q kabi yozamiz.

1-MISOL · AKVARIUM

Kichik akvariumda n ta baliq bor. Katta akvariumda undan 4 marta ko'p baliq bor. Katta akvariumda nechta baliq bor?



1. 4 marta (4 baravar) ko'p: $n + n + n + n = 4 \cdot n$.
2. Katta akvariumda $4n$ ta baliq bor. Masalan, $n = 6$ bo'lsa, $4 \cdot 6 = 24$ ta.

2-MISOL · ZAL

Zalda 9 qator stul bor, har qatorda r ta stul. Zalda nechta stul bor?

1. 1 qatorda: $1 \cdot r = r$ ta stul.
2. 9 qatorda: $9 \cdot r = 9r$ ta stul.

$t : 5$ ni kasr ko'rinishida ham yozaveramiz.



USUL · BO'LISH

t ni 5 ta teng bo'lakka bo'lamiz: $t : 5 = \dfrac{t}{5}$. Har bir bo'lak — t ning $\dfrac{1}{5}$ qismi.

24 ta konfet n ta bolaga teng bo'linsa, har biriga $24 : n = \dfrac{24}{n}$ ta tegadi.

1-MISOL · LENTA

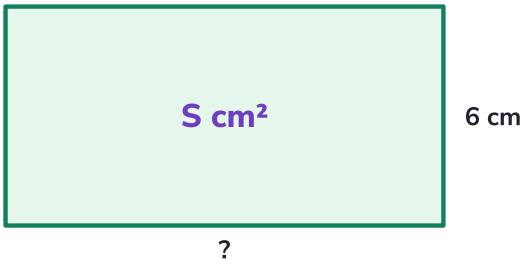
60 cm lenta 4 ta teng bo'lakka kesilsa, har bo'lak $60 : 4 = 15$ cm. L cm lenta 4 ta teng bo'lakka kesilsa, har bo'lak necha cm?



1. $L : 4 = \dfrac{L}{4}$.
2. Har bo'lak $\dfrac{L}{4}$ cm.

2-MISOL · TO'G'RI TO'RTBURCHAK

To'g'ri to'rtburchakning yuzi S cm², eni 6 cm. Bo'yi necha cm?



1. Bo'yi = yuzi : eni = $S : 6$.
2. Bo'yi $\dfrac{S}{6}$ cm.

2-QISM

Misollar ishlash

1-MISOL · BOG'

Bog'da 40 tup daraxt bor edi. Bog'bon 5 kun davomida har kuni t tupdan ko'chat o'tqazdi. Endi bog'da necha tup daraxt bor?

1. 5 kunda o'tqazilgan ko'chatlar: $5 \cdot t = 5t$ tup.
2. Ularni avvalgi 40 tupga qo'shamiz: $(40 + 5t)$ tup.

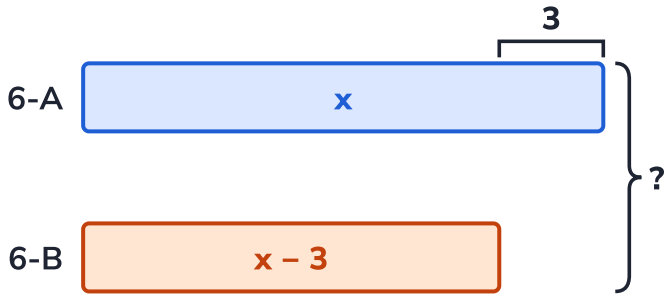
2-MISOL · BOCHKA

Bochkadagi v litr suv 4 ta chelakka teng quyildi. Keyin bitta chelakdan 3 l suv to'kildi. Shu chelakda necha litr suv qoldi?

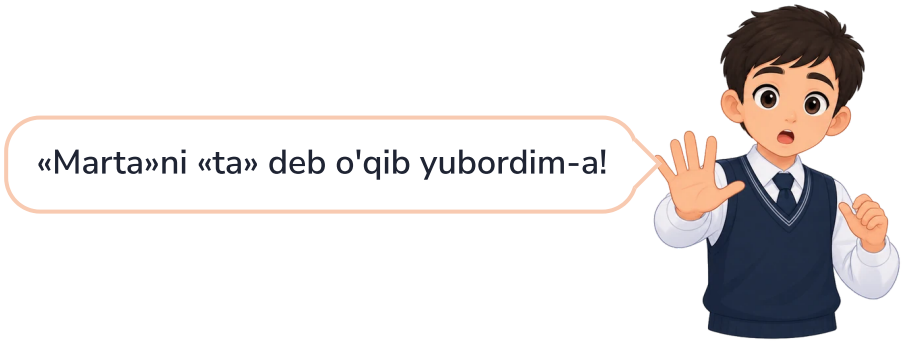
1. Bitta chelakdagi suv: $v : 4 = \frac{v}{4}$ l.
2. Undan 3 l ayiramiz: $\left(\frac{v}{4} - 3\right)$ l.

3-MISOL · IKKI SINF

6-A sinfda x nafar o'quvchi bor. 6-B sinfda undan 3 nafar kam o'quvchi bor. Ikkala sinfda jami nechta o'quvchi bor?



1. 6-B sinfda: $x - 3$ nafar.
2. Ikkala sinfni qo'shamiz: $x + (x - 3)$ nafar.



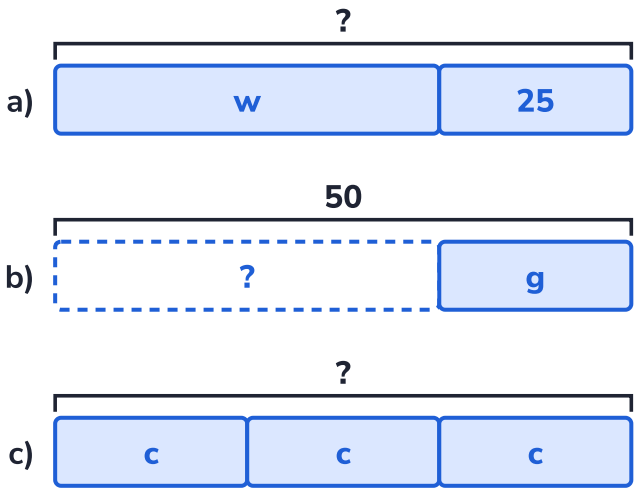
DIQQAT!

1. «m dan 8 ta ko'p»:
   ✗ $8m$ → ✓ $m + 8$. $8m$ — «8 marta ko'p».
2. «50 dan k ni ayiring»:
   ✗ $k - 50$ → ✓ $50 - k$
3. $n \cdot 7$ ni qisqa yozish:
   ✗ $n7$ → ✓ $7n$ (son oldin yoziladi)
4. «h ni 6 ga bo'ling»:
   ✗ $\frac{6}{h}$ → ✓ $\frac{h}{6}$
5. $y + 4$ ni qisqartirish:
   ✗ $y + 4 = 4y$ → ✓ $y + 4$ **shundayligicha qoladi**. $y = 2$ da $y + 4 = 6$, $4y = 8$.

3-QISM

Mustahkamlash

1. Jumlani ifoda bilan yozing: a) s ga 11 ni qo'shing; b) 27 dan j ga ortiq; c) 60 dan k ni ayiring; d) f dan 14 ga kam.
2. Algebraik ifoda yozing: a) 9 ta d; b) e ning to'rt baravari; c) m ni 5 ga bo'ling; d) 15 va b ning ko'paytmasi.
3. Chizma-modelga qarab «?» uchun ifoda yozing.



4. Javonda h ta kitob bor. Agar a) yana 6 ta kitob qo'yilsa; b) 4 ta kitob olinsa; c) kitoblar 2 marta ko'paysa, javonda nechta kitob bo'ladi?
5. Juftlikdagi ifodalar bir xilmi? a) $9 - k$ va $k - 9$; b) $\frac{k}{9}$ va $\frac{9}{k}$; c) $k \cdot 9$ va $9k$; d) $9 + k$ va $k + 9$.
6. Shokolad r so'm turadi. Muzqaymoq shokoladdan 3000 so'm arzon. Muzqaymoq qancha turadi?
7. Bog'dagi chinor hozir 16 yoshda. U necha yoshda bo'ladi yoki bo'lgan: a) n yildan keyin; b) 6 yil oldin; c) k yil oldin?
8. Kassaga 50 000 so'm berildi, xarid z so'm turadi. Qancha qaytim beriladi?
9. Velosipedchi d km yo'lni 3 kunda bosib o'tdi, har kuni bir xil masofa yurdi. U bir kunda necha km yurgan?
10. Kvadratning perimetri P cm. Uning tomoni necha cm?
11. Ikki amalli ifoda tuzing: a) k ni 2 ga bo'lib, natijaga 10 ni qo'shing; b) m ni 3 ga ko'paytirib, natijadan 4 ni ayiring.

Matnli masalalar

1. Bog'da 3 qator olma daraxti bor, har qatorda a tup. Bundan tashqari, bog'da 12 tup nok bor. Bog'da jami necha tup daraxt bor?
2. Mashina birinchi kuni s km yurdi. Ikkinchi kuni birinchi kundagi yo'lning yarmini *(yarmi — 2 ga bo'lish)*, uchinchi kuni esa ikkinchi kundagidan 15 km ko'p yurdi. Uchinchi kuni necha km yurgan?
3. Birinchi idishda v litr suv bor, ikkinchi idishda undan 3 marta ko'p. Ikkala idishda jami necha litr suv bor?
4. Sport zalida a ta to'p bor edi. 6 ta yangi to'p keltirildi, keyin 4 tasi maktab hovlisiga olib chiqildi. Zalda nechta to'p qoldi?

4-QISM

Mustaqil ish

1. Har bir jumlaga mos ifodani yozing: a) z ning olti baravari; b) 75 dan u ni ayiring; c) q ni 11 ga bo'ling; d) c dan 20 ga ortiq.
2. «h dan 7 ga kam» uchun chizma-model chizing va ifoda yozing.
3. **Xatoni toping.** Bekzod «p dan 6 marta ko'p» ni $p + 6$, «p dan 6 ta ko'p» ni esa $6p$ deb yozdi. To'g'ri ifodalarni yozing.
4. Ruchka r so'm turadi. Daftar ruchkadan 2000 so'm qimmat. Ruchka va daftar birga qancha turadi?
5. Bolalar bog'chasiga b ta o'yinchoq keltirildi va 3 ta guruhga teng bo'lindi. Keyin har bir guruhga yana 4 tadan o'yinchoq berildi. Har bir guruhda nechta o'yinchoq bo'ldi?
6. Bog'bon w tup ko'chatni o'z bog'iga va yana 2 ta qo'shni bog'ga teng bo'lib o'tqazdi. Har bir bog'ga necha tup ko'chat tegdi?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $s + 11$; b) $27 + j$; c) $60 - k$; d) $f - 14$.

2-masala

a) $9d$; b) $4e$; c) $\frac{m}{5}$; d) $15b$.

3-masala

a) $w + 25$; b) $50 - g$; c) $c + c + c = 3c$.

4-masala

a) $(h + 6)$ ta; b) $(h - 4)$ ta; c) $2h$ ta.

5-masala

a) yo'q, $9 - k$ va $k - 9$ turlicha; b) yo'q: $k = 3$ da $\frac{3}{9} = \frac{1}{3}$, $\frac{9}{3} = 3$; c) ha, tartib farqi yo'q; d) ha, tartib farqi yo'q.

6-masala

3000 so'm arzon, demak ayiramiz: $(r - 3000)$ so'm.

7-masala

a) $16 + n$; b) $16 - 6 = 10$ yoshda; c) $16 - k$.

8-masala

$(50\,000 - z)$ so'm.

9-masala

$d : 3 = \frac{d}{3}$ km.

10-masala

4 ta teng tomon: $P : 4 = \frac{P}{4}$ cm.

11-masala

a) $\frac{k}{2} + 10$; b) $3m - 4$.

Matnli masalalar

1-masala

Olma: $3a$ tup. Jami: $(3a + 12)$ tup.

2-masala

Ikkinchi kuni: $\frac{s}{2}$ km. Uchinchi kuni: $\left(\frac{s}{2} + 15\right)$ km.

3-masala

Ikkinchi idishda $3v$ l. Jami: $(v + 3v)$ l, ya'ni $4v$ l.

4-masala

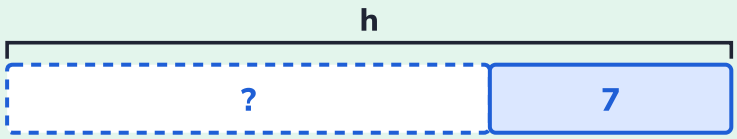
$(a + 6 - 4)$ ta, ya'ni $(a + 2)$ ta to'p.

Mustaqil ish

1-masala

a) $6z$; b) $75 - u$; c) $\frac{q}{11}$; d) $c + 20$.

2-masala



Ifoda: $h - 7$.

3-masala

Bekzod ikkalasini almashtirib yuborgan. «6 marta ko'p» — $6p$, «6 ta ko'p» — $p + 6$.

4-masala

Daftar: $(r + 2000)$ so'm. Birga: $r + (r + 2000)$ so'm.

5-masala

Bir guruhga avval $\frac{b}{3}$ ta. Keyin: $\left(\frac{b}{3} + 4\right)$ ta o'yinchoq.

6-masala

Bog'lar 3 ta: o'zining bog'i va 2 ta qo'shni bog'. Har biriga $w : 3 = \frac{w}{3}$ tup.